COURSE DESCRIPTION			
Program	**Automobile Engineering**		
Course Title	**Basic Electrical and Electronics Lab**		
Course Code	**2058**	**Semester**	**II**
Type of Course	**Lab**	**Contact Hours**	**45**
Teaching Scheme (L: T:P:R)	**0:0:3:0**	**Credits**	**1.5**
CIE Marks	**60**	**ESE Marks**	**40**

Introduction

Modern vehicles rely heavily on integrated electrical and electronic systems. This lab course provides hands-on training in using diagnostic tools, testing electrical machines, and assembling electronic circuits. It bridges the gap between theoretical concepts and practical automotive applications, ensuring students can safely maintain and troubleshoot vehicle electrical systems.

Course Objectives

1	To demonstrate the safe use of electrical tools and measurement of DC/AC quantities.
2	To conduct performance tests on DC motors, induction motors, and transformers.
3	To evaluate the health and charging characteristics of automotive batteries and alternators.
4	To construct and verify basic electronic switching and logic circuits used in automotive automation.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basics physics	2002A	Engineering Physics for Mechanical, Structural and Industrial Applications	I
Basic mathematics	1002	Fundamentals of Engineering Mathematics	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Use electrical tools and multimeters to measure parameters and verify DC/AC circuit laws.	Apply
CO2	Practice operational tests on DC motors, AC motors, and transformers to determine their characteristics.	Apply
CO3	Examine automotive battery health and charging system performance using standard testing procedures.	Apply
CO4	Exercise and test semiconductor switching circuits and digital logic gates for automation applications.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	3	2	-	-	-	2	2	-
CO2	3	2	2	3	2	-	-	-	2	2	-
CO3	3	3	2	3	3	-	2	-	2	2	2
CO4	3	2	2	3	3	-	-	-	2	2	2
Total	12	9	6	12	10	-	2	-	8	8	4
Correlation Level	3	2.25	2	3	2.5	-	2	-	2	2	2
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	4.5	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Electrical Safety & Tools	Demonstrate the safe use of insulated tools, line testers, and personal protective equipment (PPE).	CO1	Apply	2
2	Study of Digital Multimeter (DMM) and Basic Tools	Identify DMM functions (Voltage, Current, Resistance, Continuity) and use crimping tools, wire strippers, and testers.	CO1	Apply	2
3	Verification of Ohm's Law	Relate voltage, current, and resistance in a DC circuit.	CO1	Apply	2
4	Series & Parallel Circuits	Complete the construction of series and parallel resistive circuits and measure equivalent resistance.	CO1	Apply	2
5	AC Quantity Measurement	Capture the RMS value and frequency of a single-phase AC supply using a multimeter	CO1	Apply	2
Module II					
6	Transformer Testing	Calculate the voltage ratio and polarity of a single-phase transformer.	CO2	Apply	2
7	DC Motor Operation	Identify the terminals of a DC motor and complete a trial run using a starter.	CO2	Apply	2
8	DC Series Motor Control	Use a variable rheostat to control the speed of a DC series motor.	CO2	Apply	2
9	3-Phase Induction Motor	Demonstrate the construction and operation of a 3-phase squirrel cage induction motor.	CO2	Apply	3
10	Auto-transformer Study	Operate an auto-transformer to demonstrate variable AC voltage control.	CO2	Apply	2
Module III					
11	Battery Testing	Capture the terminal voltage and calculate internal resistance.	CO3	Apply	1.5
12	Battery Voltage Drop Under Load	Examine the terminal voltage variation with increasing electrical load	CO3	Apply	1.5
13	Alternator Inspection	Demonstrate the internal components of an automotive alternator (stator, rotor, and rectifiers).	CO3	Apply	3
14	Charging System Setup	Construct a basic charging circuit and monitor the charging current flow.	CO3	Apply	3
15	Electric Heating Principle	Demonstrate the principle of resistance heating using a heating element.	CO3	Apply	2
Module IV					
16	Capacitor Charging	Exercise the charging and discharging cycle of a capacitor using a DC source and a timer.	CO4	Apply	2
17	Rectifier Circuit	Construct a Bridge Rectifier circuit and verify the conversion of AC to DC.	CO4	Apply	2
18	SCR Triggering	Construct a basic circuit to trigger a Silicon Controlled Rectifier (SCR) for power control.	CO4	Apply	2
19	Transistor Switching	Construct a switching circuit using a Transistor (or MOSFET) to control a small load.	CO4	Apply	2
20	Logic Gate Verification	Practice truth tables of basic logic gates (AND, OR, NOT) using Integrated Circuits (ICs).	CO4	Apply	2
21	Sensor Interfacing	Capture the output signal of an automotive sensor (e.g., Temperature or Map sensor).	CO4	Apply	2

Open-ended experiment

Objective:
To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:
At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.
Students may choose to: Modify or extend an existing experiment to explore additional parameters or behaviors. Integrate multiple concepts from the syllabus into a combined application. Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem. Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
	Module I	
1	Identify and Trace the main fuse box and battery connections in a laboratory vehicle.	2
2	Measure the voltage of various household or automotive cells to verify series/parallel addition.	2
3	Draw a wiring diagram for a simple automotive lighting circuit (Headlamp & Switch).	1.5
	Module II	
5	Inspect the nameplate of an industrial motor and calculate its full-load current.	2
6	Compare the physical size and winding types of a starter motor vs. a wiper motor.	2
7	Simulate a transformer turns ratio calculation using an online virtual lab.	1.5
	Module III	
9	Test the specific gravity of a lead-acid battery using a hydrometer (Safety demonstration).	2
10	Sketch the internal layout of an alternator based on a cut-section model.	2
11	Research and List the heating methods used in automotive paint drying booths.	1.5
	Module IV	
13	Decode the color bands of 10 different resistors to calculate their resistance values.	2
14	Construct a simple truth table for a vehicle "Door-Open" buzzer logic.	2
15	Identify and Categorize five electronic sensors found under the hood of a modern car.	1.5
	Total Self-Learning hours	22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Fundamentals of Electrical Engineering and Electronics	B.L. Theraja	S. Chand Publications
2	Electronic Devices and Circuits	G.K. Mittal	Khanna Publishers
3	Basic Electrical and Electronics Engineering	S.K. Bhattacharya	Pearson Education

REFERENCE			
Sl. No.	Title	Author	Publication
1	Automobile Electrical and Electronic Systems	Tom Denton	Routledge
2	Automotive Electrical and Electronics	A.K. Babu	Khanna Publishing House
3	Electrical Technology (Vol. I & II)	B.L. Theraja & A.K. Theraja	S. Chand Publications
4	Basic Electrical Engineering	V.K. Mehta & Rohit Mehta	S. Chand Publications

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://nptel.ac.in/courses/108104139	I & II
2	https://www.allaboutcircuits.com/textbook/direct-current/	I
3	https://batteryuniversity.com/	III

4	https://nptel.ac.in/courses/108105053	IV
5	https://www.google.com/search?q=https://www.electronicshub.org/digital-logic-gates/	IV

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Digital Multimeter	Auto-ranging, AC/DC V/A, Resistance	10
2	DC Power Supply	Variable 0-30V, 5A	5
3	Single Phase Transformer	1kVA, 230V/115V	2
4	DC Shunt/Series Motor	0.5 HP with Starter	1 each
5	3-Phase Induction Motor	1 HP, Squirrel Cage with DOL Starter	1
6	Auto-transformer (Variac)	0-270V, 5A	2
7	Lead-Acid Battery	12V, 35Ah or higher	2
8	Battery Load Tester	High rate discharge type (12V)	1
9	Battery Load Tester	12V with external/internal regulator	2
10	Rheostats	Various ranges (50Ω, 100Ω)	5
11	Digital Oscilloscope	Dual Channel, 20MHz	2
12	Logic Trainer Kit	Built-in DC supply and logic switches	5
13	Breadboards & Components	Diodes, Transistors, SCRs, MOSFETs, ICs	15 sets
14	Automotive Sensors	NTC Thermistor, MAP sensor, LDR	5 sets
15	Electric Heating Demo	Induction cooktop or Resistance coil	1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity

Assessment Methodology – Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to			15	15	10	
Marks	5	30	15		10	40
Total marks	60					40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student’s ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50